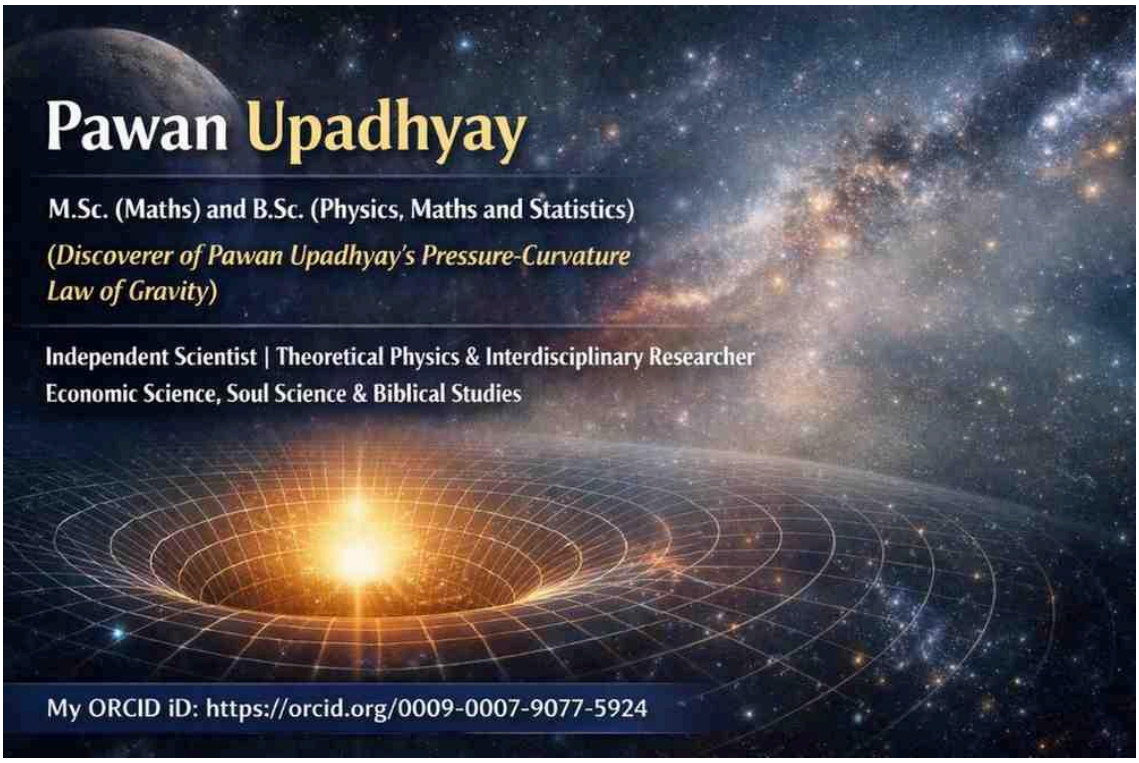
PAWAN UPADHYAY'S PRESSURE–CURVATURE LAW OF GRAVITY (PPC LAW): MAXIMUM GRAVITATIONAL PRESSURE, PERFECT-FLUID MOMENTUM CONSERVATION, AND RELATIVISTIC ACCELERATION NEAR BLACK HOLES

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ABSTRACT

Black holes represent some of the most extreme gravitational environments in the universe. Classical general relativity predicts singular behaviour in certain idealized black hole solutions, where spacetime curvature quantities become unbounded. These mathematical predictions raise fundamental questions about the physical limits of gravitational pressure, total energy density, total mass-equivalent density, and spacetime curvature.

This research paper presents an integrated theoretical framework based on Pawan Upadhyay's Pressure–Curvature Law of Gravity (PPC Law). It combines the maximum-limit principle with the PPC perfect-fluid energy-momentum tensor, relativistic momentum conservation, and pressure-gradient-driven acceleration.

The central proposition of the PPC Law is that extreme gravitational conditions are governed by maximum physical limiting values rather than physical infinities.

The fundamental PPC equations are:

$P_g = \omega E_d$

$E_d = \rho c^2$

Where:

- P_g = Gravitational pressure.
- ω = Equation-of-state parameter.
- E_d = Total energy density.
- ρ = Total mass-equivalent density.
- c = Speed of light in vacuum.

The combined PPC equation is:

$P_g = \omega \rho c^2$

The proposed limiting conditions are:

$\rho \rightarrow \rho_{max}$

$E_d \rightarrow E_{max}$

$\omega \rightarrow 1$

$P_g \rightarrow P_{g(max)}$

Consequently:

$P_{g(max)} = \rho_{max} c^2$

$E_{max} = \rho_{max} c^2$

Therefore:

$P_{g(max)} = E_{max}$

The paper incorporates the proposed PPC perfect-fluid energy-momentum tensor:

$T_{\mu\nu(PPC)} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

It also incorporates covariant conservation of the energy-momentum tensor and the relativistic acceleration equation arising from pressure gradients.

The conservation equation is:

$\nabla_\mu T^{\mu\nu(PPC)} = 0$

The general relativistic momentum-conservation equation is:

$(E_d + P_g)a^\mu = -(g^{\mu\nu} + u^\mu u^\nu)\nabla_\nu P_g$

The proposed maximum limiting condition $\omega \rightarrow 1$ corresponds to a stiff equation of state, in which gravitational pressure approaches total energy density.

This research investigates the potential application of the PPC framework to black hole interiors, gravitational collapse, relativistic fluids, and the singularity problem.

Keywords: Pawan Upadhyay, Pressure–Curvature Law, PPC Law of Gravity, Black Holes, Perfect Fluid, Energy-Momentum Tensor, Momentum Conservation, Relativistic Acceleration, Equation of State, Maximum Gravitational Pressure, Maximum Energy Density, Total Mass-Equivalent Density, Spacetime Curvature, Gravitational Singularity.

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1. INTRODUCTION

1.1. Gravity and Spacetime Curvature

Gravity governs the motion and physical evolution of astronomical objects throughout the universe.

Einstein's general theory of relativity describes gravitational phenomena through the curvature of spacetime. Matter and energy influence spacetime geometry, while the resulting geometry determines the motion of matter and radiation.

1.2. Black Holes and Extreme Gravitational Conditions

Black holes provide particularly important environments for studying extreme gravitational conditions.

During gravitational collapse, matter and energy can become concentrated within a region bounded by an event horizon.

Certain classical solutions of general relativity predict central singularities where curvature quantities become unbounded.

1.3. Central Research Question

Must extreme gravitational conditions necessarily produce physical infinities, or can gravitational pressure, total energy density, and total mass-equivalent density approach maximum physical limiting values?

Pawan Upadhyay's Pressure–Curvature Law of Gravity proposes investigating the second possibility.

1.4. Research Objective

The PPC Law introduces a maximum-limit framework in which gravitational pressure, total energy density, and total mass-equivalent density approach their respective maximum values.

The theory also emphasizes the dynamical importance of pressure.

In relativistic perfect fluids, pressure contributes to the energy-momentum distribution, while pressure gradients influence fluid acceleration.

The present research integrates the maximum gravitational pressure and maximum density principles with the PPC momentum-conservation and acceleration framework.

2. FUNDAMENTAL PRINCIPLE OF THE PPC LAW

The central principle of Pawan Upadhyay's Pressure–Curvature Law of Gravity is that extreme gravitational conditions are governed by maximum physical limiting values rather than physical infinities.

The PPC Law proposes three principal maximum quantities:

- **Maximum gravitational pressure.**
- **Maximum total energy density.**
- **Maximum total mass-equivalent density.**

These quantities are mathematically connected through the equation-of-state parameter ω.

The proposed limiting conditions are:

$\rho \rightarrow \rho_{max}$

$E_d \rightarrow E_{max}$

$\omega \rightarrow 1$

$P_g \rightarrow P_{g(max)}$

These conditions establish the maximum-limit framework of the PPC Law.

3. DEFINITIONS OF FUNDAMENTAL PPC VARIABLES

3.1. Total Mass-Equivalent Density (ρ)

The symbol ρ represents total mass-equivalent density.

It does not represent only ordinary matter density or rest-mass density.

It represents the total mass-equivalent density associated with the total energy content considered by the PPC model.

Its proposed maximum limiting condition is:

$\rho \rightarrow \rho_{max}$

Where ρ_{max} represents maximum total mass-equivalent density.

3.2. Total Energy Density (E_d)

The symbol E_d represents total energy density associated with total mass-equivalent density.

Its fundamental relationship is:

$E_d = \rho c^2$

Where:

- E_d = Total energy density.
- ρ = Total mass-equivalent density.
- c = Speed of light in vacuum.

The proposed maximum limiting condition is:

$E_d \rightarrow E_{max}$

Where E_{max} represents maximum total energy density.

3.3. Equation-of-State Parameter (ω)

The symbol ω represents the equation-of-state parameter.

Its mathematical relationship is:

$\omega = P_g/E_d$

Therefore:

$P_g = \omega E_d$

The proposed PPC limiting condition is:

$\omega \rightarrow 1$

3.4. Gravitational Pressure (P_g)

The symbol P_g represents gravitational pressure in the PPC formulation.

Its relationship with total energy density is:

$P_g = \omega E_d$

The proposed maximum limiting value is:

$P_g \rightarrow P_{g(max)}$

Where $P_{g(max)}$ represents maximum gravitational pressure.

4. FUNDAMENTAL MATHEMATICAL EQUATIONS

4.1. PPC Equation of State

$P_g = \omega E_d$

This equation establishes the relationship between gravitational pressure and total energy density.

The equation-of-state parameter determines the proportionality between these quantities.

4.2. Total Energy Density Equation

$E_d = \rho c^2$

This equation establishes the relationship between total energy density and total mass-equivalent density.

Both quantities represent the same total energy content under the PPC energy-accounting convention.

4.3. Combined PPC Equation

Starting with:

$P_g = \omega E_d$

And:

$E_d = \rho c^2$

Substituting:

$P_g = \omega (\rho c^2)$

Therefore:

$P_g = \omega \rho c^2$

This is the combined mathematical relationship of the PPC formulation.

5. MAXIMUM LIMITING CONDITIONS

5.1. Maximum Total Mass-Equivalent Density

$\rho \rightarrow \rho_{max}$

The proposed principle is that total mass-equivalent density approaches its maximum physical limiting value.

5.2. Maximum Total Energy Density

$E_d \rightarrow E_{max}$

Using:

$E_d = \rho c^2$

The maximum energy density becomes:

$E_{max} = \rho_{max} c^2$

5.3. Equation-of-State Parameter Approaches Unity

$\omega \rightarrow 1$

Using:

$P_g = \omega E_d$

We obtain:

$P_g \rightarrow E_d$

At the maximum limiting condition:

$P_{g(max)} = E_{max}$

5.4. Maximum Gravitational Pressure

$P_g \rightarrow P_{g(max)}$

The proposed principle is that gravitational pressure approaches a maximum physical limiting value rather than becoming physically infinite.

6. MATHEMATICAL DERIVATION OF MAXIMUM GRAVITATIONAL PRESSURE

Starting with:

$P_g = \omega E_d$

And:

$E_d = \rho c^2$

Combining both equations:

$P_g = \omega \rho c^2$

Applying the proposed limiting conditions:

$\rho \rightarrow \rho_{max}$

$\omega \rightarrow 1$

Therefore:

$P_{g(max)} = 1 \times \rho_{max} c^2$

Hence:

$P_{g(max)} = \rho_{max} c^2$

Similarly:

$E_{max} = \rho_{max} c^2$

Comparing both results:

$P_{g(max)} = \rho_{max} c^2$

$E_{max} = \rho_{max} c^2$

Therefore:

$P_{g(max)} = E_{max}$

This is the central maximum-pressure relationship of the proposed PPC formulation.

Pressure and energy density have identical SI dimensions:

$J/m^3 = Pa$

Therefore, the proposed equality is dimensionally consistent.

7. PPC PERFECT-FLUID ENERGY-MOMENTUM TENSOR

The PPC Law introduces a perfect-fluid energy-momentum tensor.

The proposed PPC tensor is:

$T_{\mu\nu(PPC)} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

Where:

- $T_{\mu\nu(PPC)}$ = PPC energy-momentum tensor.
- E_d = Total energy density.
- P_g = Gravitational pressure.
- u_μ = Four-velocity covector.
- u_ν = Four-velocity covector.
- $g_{\mu\nu}$ = Spacetime metric tensor.

This equation describes the energy-momentum distribution of the PPC perfect-fluid system.

7.1. Incorporating the PPC Equation of State

The equation of state is:

$P_g = \omega E_d$

Substituting into the perfect-fluid tensor:

$T_{\mu\nu(PPC)} = (E_d + \omega E_d)u_\mu u_\nu + \omega E_d g_{\mu\nu}$

Taking E_d as a common factor:

T_μv(PPC) = E_d[(1 + ω)u_μu_v + ω g_μv]

7.2. Incorporating Total Mass-Equivalent Density

Using:

E_d = ρ c²

The combined tensor becomes:

T_μv(PPC) = ρ c²[(1 + ω)u_μu_v + ω g_μv]

This equation connects the PPC energy-momentum tensor with total mass-equivalent density and the equation-of-state parameter.

7.3. PPC Tensor Under the Maximum Limiting Condition

The proposed limiting conditions are:

ρ → ρ_max

E_d → E_max

ω → 1

P_g → P_g(max)

Substituting ω = 1:

T_μv(PPC(max)) = E_max[(1 + 1)u_μu_v + g_μv]

Therefore:

T_μv(PPC(max)) = E_max(2u_μu_v + g_μv)

This represents the limiting form of the proposed PPC perfect-fluid tensor.

8. CONSERVATION OF THE PPC ENERGY-MOMENTUM TENSOR

Momentum conservation is a fundamental requirement of relativistic gravitational physics.

In general relativity, the conservation of energy and momentum is represented through covariant conservation of the stress-energy tensor.

The PPC conservation equation is:

$\nabla_\mu T^{\mu\nu}(PPC) = 0$

Where:

- ∇_μ = Covariant derivative.
- $T^{\mu\nu}(PPC)$ = Contravariant PPC energy-momentum tensor.

This equation expresses local energy-momentum conservation for the PPC perfect-fluid system when it is treated as a separately conserved source.

8.1. Physical Significance of Momentum Conservation

The PPC framework investigates the relationship between:

- Total energy density.
- Gravitational pressure.
- Fluid motion.
- Pressure gradients.
- Relativistic acceleration.
- Spacetime curvature.

The conservation equation connects these physical quantities through relativistic fluid dynamics.

8.2. Relationship With Einstein's Field Equations

In standard general relativity, Einstein's field equations are:

$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$

Where:

- $G_{\mu\nu}$ = Einstein tensor.
- G = Newton's gravitational constant.
- c = Speed of light.
- $T_{\mu\nu}$ = Energy-momentum tensor.

The contracted Bianchi identity implies:

$\nabla_\mu G^{\mu\nu} = 0$

Consequently:

$\nabla_\mu T^{\mu\nu} = 0$

The PPC perfect-fluid tensor satisfies the corresponding conservation requirement when used as a separately conserved source in standard general relativity.

9. GENERAL RELATIVISTIC FORM OF PPC MOMENTUM CONSERVATION AND ACCELERATION

The PPC framework incorporates the general relativistic momentum-conservation equation.

The original PPC equation is:

$(E_d + P_g)a^\mu = -(g^{\mu v} + u^\mu u^v)\nabla_v P_g$

This equation must retain the original spacetime indices μ and v.

Definitions:

- E_d = Total energy density.
- P_g = Gravitational pressure.
- a^μ = Four-acceleration.
- $g^{\mu v}$ = Contravariant spacetime metric tensor.
- u^μ = Contravariant four-velocity.
- u^v = Four-velocity component.
- ∇_v = Covariant derivative.
- μ and v = Spacetime indices.

9.1. Four-Acceleration

The four-acceleration is defined as:

$a^\mu = u^v \nabla_v u^\mu$

It describes the change in four-velocity along the fluid's worldline.

9.2. Projection Tensor

Define the projection tensor:

$h^{\mu v} = g^{\mu v} + u^\mu u^v$

For metric signature (-+++) and normalized four-velocity $u^\mu u_\mu = -1$, this tensor projects quantities perpendicular to the fluid four-velocity.

The PPC momentum equation can therefore be expressed as:

$(E_d + P_g)a^\mu = -h^{\mu v}\nabla_v P_g$

9.3. Relativistic Acceleration Equation

Rearranging:

a^μ = -h^μv∇_vP_g/(E_d + P_g)

Expanding the projection tensor:

a^μ = -(g^μv + u^μu^v)∇_vP_g/(E_d + P_g)

This equation describes pressure-gradient-driven relativistic acceleration in the PPC perfect-fluid framework.

9.4. Physical Interpretation

The equation establishes that:

- Pressure gradients contribute to fluid acceleration.
- Total energy density contributes to the effective inertial term.
- Gravitational pressure contributes to the effective inertial term.
- The projection tensor identifies the pressure-gradient component perpendicular to the fluid four-velocity.

Relativistic Momentum Conservation and the PPC Interpretation

1. General Relativistic Form

The local conservation of energy and momentum in curved spacetime is expressed by

(1)

$$\nabla_\mu T^{\mu\nu} = 0$$

where

- ∇_μ is the covariant derivative,
- $T^{\mu\nu}$ is the energy-momentum tensor.

For a perfect fluid,

(2)

$$T^{\mu\nu} = (E_d + P_g)u^\mu u^\nu + P_g g^{\mu\nu}$$

Projecting Equation (1) orthogonally to the four-velocity gives the relativistic Euler equation

(3)

$$(E_d + P_g)a^\mu = -\left(g^{\mu\nu} + u^\mu u^\nu\right)\nabla_\nu P_g$$

where

(4)

$$a^\mu = u^\nu \nabla_\nu u^\mu$$

is the four-acceleration.

10. INCORPORATING THE PPC EQUATION OF STATE INTO RELATIVISTIC ACCELERATION

The PPC equation of state is:

$P_g = \omega E_d$

Taking the covariant derivative:

$\nabla_v P_g = \nabla_v(\omega E_d)$

Applying the product rule:

$\nabla_v P_g = \omega \nabla_v E_d + E_d \nabla_v \omega$

Substituting into the relativistic acceleration equation:

$a^\mu = -h^{\mu v}[\omega \nabla_v E_d + E_d \nabla_v \omega]/(E_d + P_g)$

Since:

$P_g = \omega E_d$

The denominator becomes:

$E_d + P_g = E_d(1 + \omega)$

Therefore:

$a^\mu = -h^{\mu v}[\omega \nabla_v E_d + E_d \nabla_v \omega]/[E_d(1 + \omega)]$

10.1. Special Case: Constant Equation-of-State Parameter

If ω is constant:

$\nabla_v \omega = 0$

Then:

$a^\mu = -[\omega/(1 + \omega)]h^{\mu v} \nabla_v E_d/E_d$

Therefore:

$a^\mu = -[\omega/(1 + \omega)]h^{\mu v} \nabla_v \ln(E_d)$

10.2. Maximum Limiting Condition

According to the proposed PPC limiting condition:

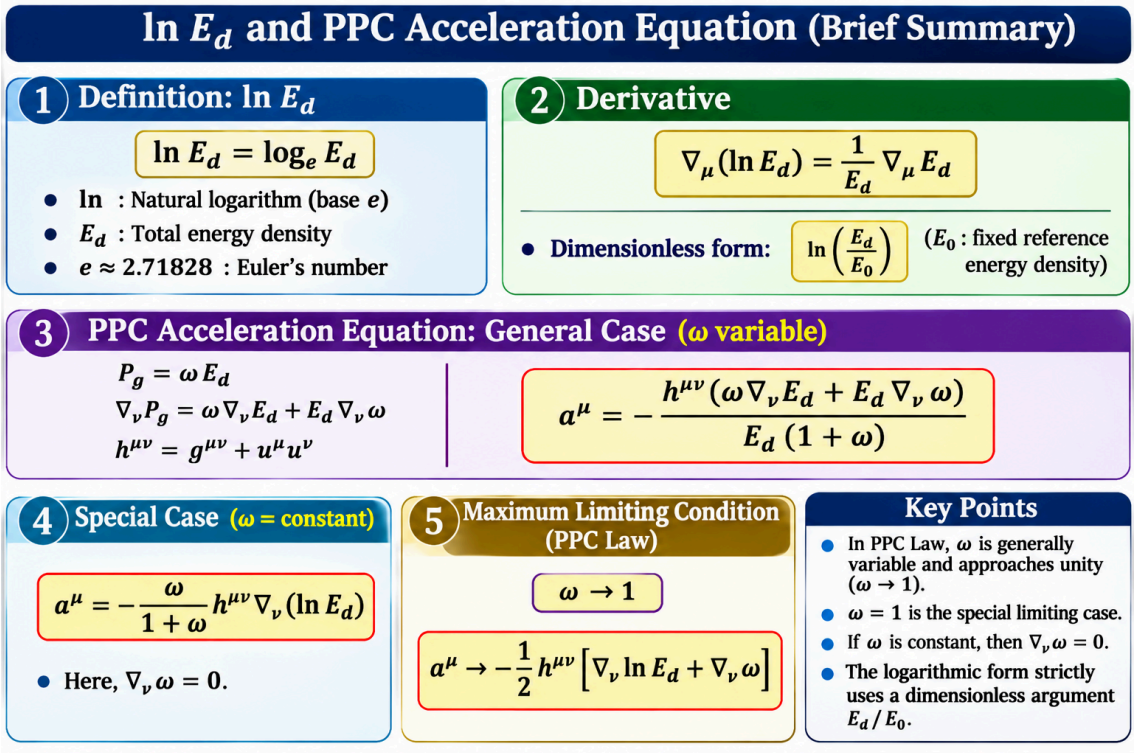
ω → 1

Substituting:

a^μ = -1/2 h^μν∇_ν ln(E_d)

This result applies when ω is constant and equal to unity in the relevant fluid region.

It describes the pressure-gradient contribution to relativistic acceleration in the proposed stiff-fluid limit.



11. ENERGY CONSERVATION EQUATION

The conservation equation also produces an energy evolution equation.

Projecting the conservation equation along the four-velocity gives:

u^μ∇_μE_d + (E_d + P_g)∇_μu^μ = 0

Define the fluid expansion:

θ = ∇_μu^μ

Then:

$u^\mu \nabla_\mu E_d + (E_d + P_g)\theta = 0$

Substituting:

$P_g = \omega E_d$

We obtain:

$u^\mu \nabla_\mu E_d + E_d(1 + \omega)\theta = 0$

Therefore:

$u^\mu \nabla_\mu E_d = -(1 + \omega)E_d\theta$

Under the proposed limiting condition:

$\omega \rightarrow 1$

The equation becomes:

$u^\mu \nabla_\mu E_d = -2E_d\theta$

This demonstrates how total energy density evolves with the expansion or contraction of the relativistic fluid.

For a contracting fluid, θ is negative, and energy density increases along the fluid flow under this idealized model.

12. MAXIMUM PRESSURE AND MOMENTUM CONSERVATION NEAR BLACK HOLES

The PPC maximum-limit principle can be considered together with the momentum-conservation equations.

The proposed limiting conditions are:

$\rho \rightarrow \rho_{max}$

$E_d \rightarrow E_{max}$

$\omega \rightarrow 1$

$P_g \rightarrow P_g(max)$

The maximum-pressure relationship is:

$P_{g(max)} = E_{max}$

The maximum-density relationship is:

$E_{max} = \rho_{max} c^2$

Consequently:

$P_{g(max)} = \rho_{max} c^2$

The PPC perfect-fluid tensor approaches:

$T_{\mu\nu(PPC(max))} = E_{max}(2u_\mu u_\nu + g_{\mu\nu})$

This provides the proposed limiting energy-momentum representation of the PPC fluid.

13. PPC INTERPRETATION OF GRAVITATIONAL PRESSURE AND SPACETIME CURVATURE

The PPC framework emphasizes the relationship between energy density, pressure, and spacetime curvature.

Its principal propositions are:

- Total energy density contributes to gravitational dynamics.
- Pressure contributes to the energy-momentum distribution.
- Pressure gradients influence relativistic fluid acceleration.
- Fluid motion takes place within curved spacetime.
- The maximum-limit conditions constrain the physical quantities considered by the PPC model.

The pressure-gradient acceleration equation is:

$a^\mu = -h^{\mu\nu}\nabla_\nu P_g/(E_d + P_g)$

The combined PPC pressure equation is:

$P_g = \omega\rho c^2$

These equations connect the pressure distribution with relativistic fluid dynamics.

14. APPLICATION TO BLACK HOLE PHYSICS

Black holes provide an important theoretical environment for investigating the PPC maximum-limit principle.

The framework proposes investigating three physical quantities:

- Maximum gravitational pressure.
- Maximum total energy density.
- Maximum total mass-equivalent density.

The proposed limiting conditions are:

$\rho \rightarrow \rho_{max}$

$E_d \rightarrow E_{max}$

$\omega \rightarrow 1$

$P_g \rightarrow P_{g(max)}$

14.1. Event Horizon

The event horizon is the boundary beyond which signals cannot escape to distant external observers.

In classical general relativity, the Schwarzschild event horizon is not itself a curvature singularity.

The PPC framework distinguishes the event horizon from the deeper gravitational conditions inside a black hole.

14.2. Black Hole Interior

The interior of a black hole provides a theoretical environment for examining the proposed maximum limits.

The PPC framework investigates whether total energy density and total mass-equivalent density can approach maximum physical values rather than becoming unbounded.

The perfect-fluid conservation equations provide a mathematical framework for studying pressure gradients and fluid evolution.

14.3. Central Gravitational Region

The central gravitational region is particularly important when investigating the classical singularity problem.

The PPC hypothesis proposes that gravitational pressure, total energy density, and total mass-equivalent density approach finite maximum limits.

The mathematical framework investigates:

- Maximum total mass-equivalent density.
- Maximum total energy density.
- Maximum gravitational pressure.
- The limiting PPC energy-momentum tensor.
- Relativistic momentum conservation.
- Pressure-gradient-driven acceleration.

15. APPLICATIONS TO NEUTRON STARS AND RELATIVISTIC FLUIDS

The PPC momentum-conservation framework can also be applied to the study of neutron stars, relativistic plasmas, early-universe cosmology, and high-energy astrophysical systems.

15.1. Neutron Stars

Neutron stars possess extremely high energy densities and internal pressures.

The PPC framework investigates:

- Pressure contributions to gravitational dynamics.
- Relativistic fluid acceleration.
- Density-dependent equation-of-state behaviour.
- Possible maximum-density conditions.
- Stability during gravitational collapse.

15.2. Relativistic Plasmas

Relativistic plasmas can exhibit substantial energy densities and pressure gradients.

The PPC conservation framework provides a mathematical basis for investigating the relationship between pressure, energy density, and fluid acceleration.

15.3. Cosmology

In relativistic cosmology, the universe can be approximated by a homogeneous and isotropic perfect fluid.

The PPC equation of state is:

$P_g = \omega E_d$

This relationship provides a mathematical framework for investigating pressure and energy-density evolution.

The proposed limiting condition $\omega \rightarrow 1$ represents a stiff-fluid regime.

16. COMPARISON WITH CLASSICAL GENERAL RELATIVITY

Classical general relativity:

- Uses Einstein's field equations to describe spacetime curvature.
- Represents perfect-fluid matter through its energy-momentum tensor.
- Requires covariant conservation of the matter energy-momentum tensor.
- Describes pressure-gradient-driven acceleration through relativistic fluid equations.
- Predicts singular behaviour in certain gravitational-collapse solutions.

Proposed PPC framework:

- Introduces maximum total mass-equivalent density.
- Introduces maximum total energy density.
- Introduces maximum gravitational pressure.
- Uses the equation-of-state relationship $P_g = \omega E_d$.
- Proposes the limiting condition $\omega \rightarrow 1$.
- Uses the PPC perfect-fluid energy-momentum tensor.
- Incorporates momentum conservation.
- Incorporates pressure-gradient-driven relativistic acceleration.
- Investigates maximum-limit behaviour near black holes.

The proposed PPC contribution is its combination of the gravitational-pressure interpretation and maximum-limit hypotheses.

17. OBSERVATIONAL AND EXPERIMENTAL RESEARCH DIRECTIONS

17.1. Gravitational Waves

Black hole mergers generate gravitational waves.

The PPC framework can be investigated for possible differences in merger or ringdown behaviour.

17.2. Black Hole Shadows

Black hole shadow observations provide information about surrounding spacetime geometry.

A complete PPC black hole solution can be used to investigate whether its exterior geometry produces distinguishable predictions.

17.3. Accretion Phenomena

Matter surrounding black holes forms accretion structures and produces radiation.

The PPC framework can investigate possible differences in accretion dynamics.

These represent proposed research directions for testing the PPC framework.

18. DISCUSSION

The integrated PPC framework combines three important theoretical elements.

FIRST: MAXIMUM-LIMIT PRINCIPLE

Gravitational pressure, total energy density, and total mass-equivalent density approach maximum physical limiting values.

SECOND: PPC PERFECT-FLUID ENERGY-MOMENTUM TENSOR

The tensor describes the proposed relationship between total energy density, gravitational pressure, and relativistic fluid motion.

THIRD: RELATIVISTIC MOMENTUM CONSERVATION

Covariant momentum conservation establishes how pressure gradients influence relativistic acceleration.

The mathematical relationships are:

$P_g = \omega E_d$

$E_d = \rho c^2$

$P_g = \omega \rho c^2$

$T_{\mu\nu}(PPC) = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

$\nabla_\mu T^{\mu\nu}(PPC) = 0$

$(E_d + P_g)a^\mu = -(g^{\mu\nu} + u^\mu u^\nu)\nabla_\nu P_g$

The proposed limiting conditions are:

$\rho \rightarrow \rho_{max}$

$E_d \rightarrow E_{max}$

$\omega \rightarrow 1$

$P_g \rightarrow P_g(max)$

The central maximum-pressure relationship is:

$P_{g(max)} = E_{max} = \rho_{max} c^2$

Together, these relationships provide the proposed mathematical framework for investigating maximum gravitational pressure, relativistic fluid motion, and bounded physical quantities near black holes.

19. CONCLUSION

Pawan Upadhyay's Pressure–Curvature Law of Gravity presents a proposed theoretical framework for investigating extreme gravitational conditions near black holes.

The central principle is that gravitational pressure, total energy density, and total mass-equivalent density approach maximum physical limiting values rather than physical infinities.

The fundamental equations are:

$P_g = \omega E_d$

$E_d = \rho c^2$

Combining them:

$P_g = \omega \rho c^2$

The proposed limiting conditions are:

$\rho \rightarrow \rho_{max}$

$E_d \rightarrow E_{max}$

$\omega \rightarrow 1$

$P_g \rightarrow P_{g(max)}$

The maximum gravitational pressure is:

$P_{g(max)} = \rho_{max} c^2$

The maximum total energy density is:

$E_{max} = \rho_{max} c^2$

Therefore:

$P_{g(max)} = E_{max}$

The PPC perfect-fluid energy-momentum tensor is:

$T_{\mu\nu}(PPC) = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

Its combined form is:

$T_{\mu\nu}(PPC) = \rho c^2[(1 + \omega)u_\mu u_\nu + \omega g_{\mu\nu}]$

At the maximum limiting condition:

$T_{\mu\nu}(PPC(max)) = E_{max}(2u_\mu u_\nu + g_{\mu\nu})$

The conservation equation is:

$\nabla_\mu T^{\mu\nu}(PPC) = 0$

The general relativistic momentum-conservation equation is:

$(E_d + P_g)a^\mu = -(g^{\mu\nu} + u^\mu u^\nu)\nabla_\nu P_g$

These equations connect the PPC maximum-limit principle with relativistic perfect-fluid dynamics and pressure-gradient-driven acceleration.

The integrated framework provides a theoretical direction for investigating gravitational collapse, black hole interiors, and the physical limits of pressure and energy density.

20. CENTRAL SCIENTIFIC PROPOSITION

Pawan Upadhyay's Pressure–Curvature Law of Gravity (PPC Law) proposes that extreme gravitational conditions near black holes are governed by maximum limiting values of gravitational pressure, total energy density, and total mass-equivalent density rather than physical infinities.

The fundamental mathematical relationships are:

$P_g = \omega E_d$

$E_d = \rho c^2$

$P_g = \omega\rho c^2$

The proposed limiting conditions are:

$\rho \rightarrow \rho_{max}$

$E_d \rightarrow E_{max}$

$\omega \rightarrow 1$

$P_g \rightarrow P_g(max)$

The resulting maximum-pressure relationship is:

$P_{g(max)} = E_{max} = \rho_{max} c^2$

The PPC perfect-fluid tensor is:

$T_{\mu\nu(PPC)} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

The conservation principle is:

$\nabla_\mu T^{\mu\nu(PPC)} = 0$

The relativistic momentum-conservation equation is:

$(E_d + P_g)a^\mu = -(g^{\mu\nu} + u^\mu u^\nu)\nabla_\nu P_g$

General Relativistic Form

$$(E_d + P_g)a^\mu = -\left(g^{\mu\nu} + u^\mu u^\nu\right)\nabla_\nu P_g$$

Together, these relationships establish the proposed mathematical framework for investigating maximum gravitational pressure, relativistic fluid motion, and bounded physical quantities near black holes.

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RESEARCH STATUS AND AUTHORSHIP

Author: Pawan Upadhyay.

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Theory: Pawan Upadhyay’s Pressure–Curvature Law of Gravity (PPC Law of Gravity).

Research Focus: Maximum gravitational pressure, maximum total energy density, maximum total mass-equivalent density, PPC perfect-fluid dynamics, momentum conservation, and relativistic acceleration near black holes.

This manuscript presents the proposed theoretical framework based on the author's PPC equations and maximum-limit principles.

The research integrates the PPC perfect-fluid tensor, equation of state, momentum-conservation relationship, and pressure-gradient-driven acceleration with the maximum-limit framework.

The PPC maximum-limit principles and their application to black hole singularities are proposed theoretical results whose broader physical predictions require mathematical derivation and empirical investigation.

Point No. 11: Energy Conservation Equation in PPC Law

1 Starting Equation: PPC Energy-Momentum Conservation

$$\nabla_\mu T^{\mu\nu}_{PPC} = 0$$

PPC Perfect-Fluid Tensor:

$$T^{\mu\nu}_{PPC} = (E_d + P_g)\,u^\mu u^\nu + P_g\, g^{\mu\nu}$$

2 Energy Conservation Equation

Projecting the conservation equation along the four-velocity gives:

$$u^\mu \nabla_\mu E_d + (E_d + P_g)\nabla_\mu u^\mu = 0$$

Define fluid expansion: $\theta = \nabla_\mu u^\mu$

Therefore: $u^\mu \nabla_\mu E_d + (E_d + P_g)\,\theta = 0$

Symbols and Definitions

E_d	=	Total energy density
P_g	=	Gravitational pressure
u^μ	=	Four-velocity (contravariant)
u^ν	=	Four-velocity (contravariant)
$g^{\mu\nu}$	=	Spacetime metric tensor
∇_μ	=	Covariant derivative
θ	=	Fluid expansion $(\nabla_\mu u^\mu)$
ω	=	Equation-of-state parameter
ρ	=	Total mass-equivalent density
c	=	Speed of light in vacuum

3 Incorporating the PPC Equation of State

Equation of state: $P_g = \omega E_d$ $\Rightarrow$ Substituting, we get: $u^\mu \nabla_\mu E_d + E_d(1+\omega)\,\theta = 0$

Therefore: $u^\mu \nabla_\mu E_d = -(1+\omega)\,E_d\,\theta$

4 Maximum Limiting Condition

According to PPC Law: $\omega \to 1$ $\Rightarrow$ Substituting, we get: $u^\mu \nabla_\mu E_d = -2\,E_d\,\theta$

Since: $E_d = \rho c^2$

For constant c: $u^\mu \nabla_\mu \rho = -2\,\rho\,\theta$

5 Physical Interpretation

- **Expansion:** When θ is positive $(\theta > 0)$, total energy density decreases along the fluid flow.
- **Contraction:** When θ is negative $(\theta < 0)$, total energy density increases along the fluid flow.
- **Maximum Limiting Condition:** When ω approaches unity $(\omega \to 1)$, the energy-density evolution equation takes the form: $u^\mu \nabla_\mu E_d = -2\,E_d\,\theta$.

$$u^\mu \nabla_\mu E_d + (E_d + P_g)\nabla_\mu u^\mu = 0$$

Point No. 11: Energy Conservation in PPC Law (Short Form)

1 PPC Perfect-Fluid Tensor

$$T^{\mu\nu}_{PPC} = (E_d + P_g)u^\mu u^\nu + P_g\, g^{\mu\nu}$$

2 Conservation and Contraction

$$\nabla_\mu T^{\mu\nu}_{PPC} = 0 \quad \Rightarrow \quad u_\nu \nabla_\mu T^{\mu\nu}_{PPC} = 0$$

3 Energy Conservation Equation

$$u^\mu \nabla_\mu E_d + (E_d + P_g)\, \nabla_\mu u^\mu = 0$$

Define fluid expansion: $\theta = \nabla_\mu u^\mu$

$$\Rightarrow \quad u^\mu \nabla_\mu E_d + (E_d + P_g)\, \theta = 0$$

4 Using PPC Equation of State

$$P_g = \omega E_d \quad \Rightarrow \quad u^\mu \nabla_\mu E_d = -(1+\omega)\, E_d \theta$$

5 Maximum Limiting Condition

$$\omega \to 1 \quad \Rightarrow \quad u^\mu \nabla_\mu E_d = -2\, E_d \theta$$

$$E_d = \rho c^2 \quad \Rightarrow \quad u^\mu \nabla_\mu \rho = -2\, \rho \theta$$

Symbols

E_d	: Total energy density
P_g	: Gravitational pressure
u^μ	: Contravariant four-velocity
u_ν	: Covariant four-velocity
$g^{\mu\nu}$	: Spacetime metric tensor
∇_μ	: Covariant derivative
θ	: Fluid expansion ($\nabla_\mu u^\mu$)
ω	: Equation-of-state parameter
ρ	: Total mass-equivalent density
c	: Speed of light in vacuum

Physical Interpretation

- **Expansion** ($\theta > 0$): E_d decreases along the fluid flow.
- **Contraction** ($\theta < 0$): E_d increases along the fluid flow.
- $\omega \to 1$: Energy-density evolution becomes $u^\mu \nabla_\mu E_d = -2\, E_d \theta$.